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Roll No

CS-7004(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Embedded Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss various challenges in Embedded Computing system design.
b) Explain the role of Watch-dog Timer in Embedded System.
2. Write a short note on five levels of abstraction from top to bottom in design process under following heads: Requirements, Specifications, Architecture, Components and system integration.
3. a) What do you mean by debug monitor?
b) What is the difference between linker, Loader and locator in embedded systems?
4. a) What is Round-Robin software architecture? What are its advantages?
b) Explain Embedded software development tools in details.

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5. a) Explain distributed embedded system. Mention its advantages.
b) Explain with all necessary sketches to enable intra communications among peripherals using I²C bus.
6. a) With the help of a suitable example explain the concept of Semaphores.
b) How to evaluate operating system performance? Give details.
7. a) What do you mean by the term advanced configuration and power interface?
b) Write a short note on performance metrics for embedded real time operating system under following heads:- Dependencies, Measurement, Interrupt latency, Scheduling latency, Timing kernel services.
8. Write short notes on :
 - i) Inter task communication
 - ii) DMA
